Hidden in Plain Sight: Unmasking Professional Airbnb Hosts with Public Data

Quan Changhua | Xing Zhenning | Chen Yuejie

Identifying Professional Airbnb Hosts in Barcelona Using Guest-Visible Information

Abstract

This study identifies "professional hosts" (≥ 3 listings) among 4,595 Airbnb hosts in Barcelona using solely public guest-visible information. By implementing strict Host-level data isolation to prevent leakage, our Random Forest model achieved an ROC-AUC of 0.7767. Incorporating a 1:4 asymmetric cost matrix, we derived an optimal intervention threshold ($t = 0.7072$) that minimizes empirical social loss. Furthermore, unstructured text mining captured commercial proxy signals, boosting the test ROC-AUC to 0.8766 and reducing global policy loss by 22.2%. Finally, unsupervised deconstruction revealed a logical disintegration between the administrative listing-count metric and true underlying operational models. This research provides municipal regulators with a validated, cost-sensitive "Triage Tool" for law enforcement prioritization.

Keywords: Machine Learning; Short-Term Rental Regulation; Data Leakage Prevention; Cost-Sensitive Decision Making; Text Analysis

1 Introduction

The proliferation of short-term rentals has introduced negative externalities to urban housing in cities like Barcelona. Precise regulation requires distinguishing "Professional Hosts" from "Ordinary Hosts". However, severe information asymmetry restricts regulators to front-end "guest-visible information," as platforms refuse to share backend data.

To build a reliable recognition model, this study established strict "Leakage-proof" validation at the physical "Host" level, preventing cross-set leakage caused by traditional "Listing" level splits. Furthermore, we integrated a 1:4 asymmetric misclassification cost matrix to derive a Bayesian optimal policy intervention threshold, moving beyond pure algorithmic metrics.

Finally, utilizing Natural Language Processing (NLP) to extract commercial proxy signals and unsupervised learning to deconstruct market micro-structures, we confirmed that the administrative "number of listings" metric logically disintegrates from true operational models. This research provides a scientific triage tool for law enforcement and establishes a prudent Human-in-the-Loop paradigm for digital urban governance.

2 Data & Experimental Design

The empirical data for this study is drawn from a Barcelona city snapshot provided by Inside Airbnb. We imposed strict methodological constraints on target definition, feature extraction, and validation architecture.

2.1 Sample and Target Definition

The cleaned final sample contains 15,293 valid listings, belonging to 4,595 independent hosts. The predictive entity in this study is strictly aligned as the "Host" rather than a single "Listing". We defined hosts owning 3 or more listings as "Professional Hosts" (target variable $Target = 1$), and the rest as "Ordinary Hosts" ($Target = 0$). Descriptive statistics show that among the 4,595 hosts, there are 982 professional hosts, accounting for 21.35%, which constitutes a mildly imbalanced classification task. All models generate predicted probabilities at the micro-listing level, which are then aggregated by `host_id` to calculate the mean, ultimately completing the evaluation at the macro-host level.

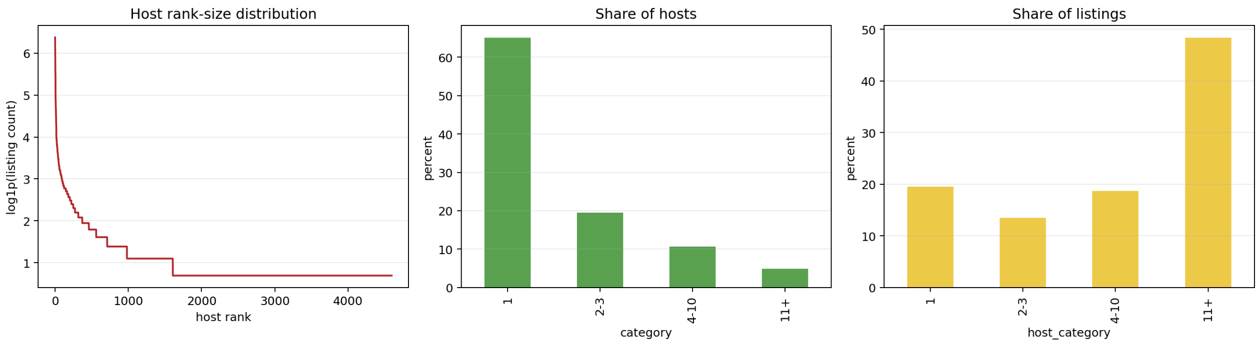

Figure 1: Distribution of Listing Holdings per Host and Binary Target Class Structure (Target ≥ 3)

As shown in Figure 1, the asset distribution of hosts in the Barcelona short-term rental market exhibits extreme long-tail right-skewness and polarization. 67.2% of hosts only list 1 property, demonstrating part-time sharing characteristics; however, a very small number of top professional hosts accumulate dozens of listings, forming a significant concentration effect. Figure 1 intuitively reveals the one-to-many mapping between hosts and listings, conclusively proving that traditional "Random Row Splits" would cause listings from the same host to scatter simultaneously across training and test sets, allowing the model to cheat by "memorizing host IDs". This laid the methodological foundation that this study must implement grouped partitioning at the physical `host_id` level.

2.2 Feature Space: Guest-Visible Constraints

To reduce data acquisition costs for regulatory departments, feature engineering set an absolute red line: only use information visible to ordinary guests on the front end. We thoroughly removed platform backend data and derived leakage fields (such as `calculated_host_listings_count`). The constructed structured feature space is divided into four dimensions. First are asset and capacity features, including log price, accommodates, number of bedrooms/bathrooms, and property category (`room_type`). Second is the operational and calendar structure, covering the minimum stay requirement and availability rates over the next 30/365 days. Third is market verification and activity, including total reviews, ratings, and platform tenure. Finally, host front-end badges, such as Superhost status and response speed. The detailed definitions and types of structured features are shown in Table 1.

2.3 Experimental Design and Zero-Leakage Mechanisms

To obtain an absolutely honest estimate of generalization error, this study constructed a triple zero-leakage validation mechanism. First, Absolute Hold-out: before any data exploration, 25% of the data was randomly and permanently frozen by host dimension as an unknown test set. Second, Host-level GroupKFold: employing a 5-fold cross-validation with `host_id` as the grouping key, ensuring that all properties of any given host fall entirely into a single partition. Third, Pipeline Encapsulation: strict encapsulation of log transformations, imputation, standardization, and one-hot encoding into a Pipeline, ensuring that the fitting of means and variances is strictly confined within each training fold, thereby blocking data leakage.

3 Supervised Learning & Decision

We decoupled the prediction task into two stages: first, estimating the conditional probability that a host is a "professional operator," and then setting an optimal cutoff threshold based on law enforcement costs.

Table 1: Guest-Visible Structured Features and Definitions

Variable Name	**Type**	**Definition**
Panel A: Asset & Capacity		
`log_price_eur`	Numeric	Natural logarithm of the nightly listing price in EUR.
`accommodates`	Numeric	Maximum number of guests the property can accommodate.
`bedrooms`	Numeric	Total number of bedrooms in the listing.
`beds`	Numeric	Total number of beds available for guests.
`bathroom_number`	Numeric	Total number of bathrooms in the listing.
`room_type`	Categorical	Property category (Entire home/apt, Private room, Shared room, Hotel room).
Panel B: Operational & Calendar		
`log_minimum_nights`	Numeric	Natural logarithm of the minimum required stay in days.
`availability_30`	Numeric	Number of available booking days within the next 30 days.
`availability_365`	Numeric	Number of available booking days within the next 365 days.
Panel C: Market Verification & Activity		
`log_number_of_reviews`	Numeric	Natural logarithm of the cumulative review count.
`has_reviews`	Binary	Indicator variable (1 if the listing has at least one review, 0 otherwise).
`review_scores_rating`	Numeric	Overall customer rating score on a 0–5 scale.
`host_tenure_months`	Numeric	Total number of months the host has been registered on the platform.
Panel D: Host Front-End Badges		
`host_is_superhost`	Categorical	Indicator for whether the host has achieved Superhost status.
`host_identity_verified`	Categorical	Indicator for whether the host's identity has been verified.
`host_response_time`	Categorical	Average response time efficiency tier of the host.

3.1 Benchmark vs. Challenger

To evaluate probability estimation performance, we built a strict benchmarking system. The benchmark model used Logistic Regression with L_1 (Lasso) and L_2 (Ridge) regularization penalties, while the challenger model introduced the Random Forest. Given the mild class imbalance of 21.35%, the evaluation abandoned accuracy and instead adopted ROC-AUC to measure global ranking ability and Average Precision (PR-AUC) to assess recognition sharpness for the minority class (professional hosts).

Figure 2: OOF Performance and Coefficient Penalization of Regularized Logistic Regression (Ridge and Lasso)

As shown in Figure 2, regularized logistic regression demonstrated stable convergence characteristics during cross-validation, effectively shrinking the weights of highly collinear calendar features. However, unable to spontaneously capture high-dimensional interaction terms, its linear expression upper limit was fixed.

As shown in Figure 3, the Random Forest model exhibited a significant advantage on the independent

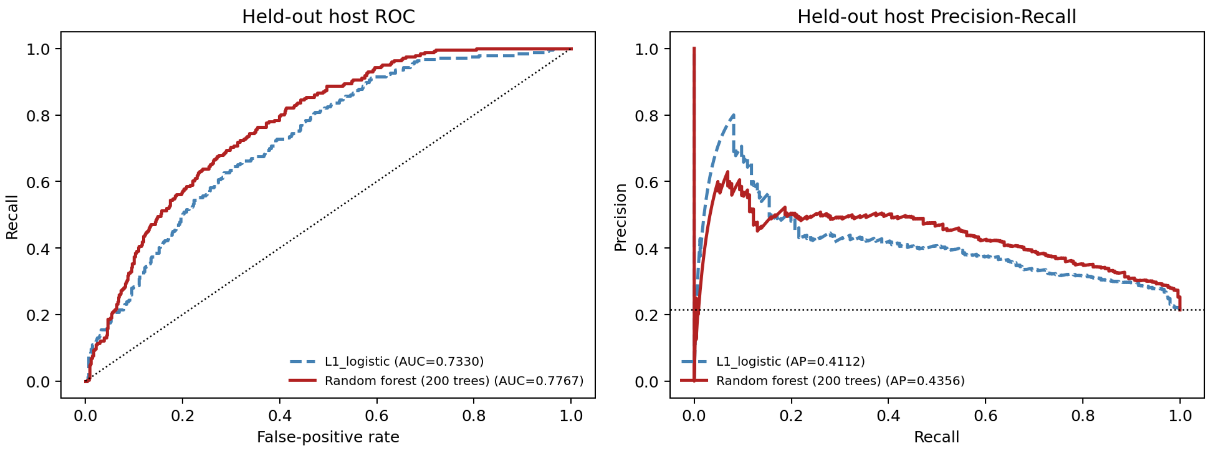

Figure 3: ROC and Precision-Recall Curves for the Random Forest Classifier on Hold-out Test Set

test set, achieving an ROC-AUC of 0.7767 and a PR-AUC of 0.4356. The high alignment between the test set curves and the Out-of-Fold (OOF) development set trajectories confirmed that the host-level isolation mechanism successfully prevented model overfitting.

3.2 Non-linear Interactions & Feature Decoding

The fundamental reason Random Forest outperformed the linear model lies in the decision tree's natural ability to capture non-linearities and interactions. To restore economic interpretability, we employed Permutation Importance to decode the features. The analysis indicated that the implicit interaction between the "calendar vacancy structure" (e.g., `availability_365`) and "property category" (`room_type`) occupied a core position. When the property category is "Entire home/apt", the minimum stay is extremely short, and the future calendar is highly open, the asset exhibits strong "Hotelification" characteristics, greatly driving up the predicted probability of being a professional host.

3.3 Cost-Sensitive Decision Making

In actual regulation, the default 0.5 threshold implies the assumption of equal misclassification costs. In short-term rental governance, the social cost of missing a professional host (False Negative, FN), which leads to long-term rental loss and rent hikes, is far higher than the administrative inspection cost of wrongfully investigating an ordinary host (False Positive, FP). We set an asymmetric cost ratio of $C_{FP} : C_{FN} = 1 : 4$. The empirical total social loss function $\mathcal{L}(t)$ is expressed as:

$$\mathcal{L}(t) = \sum_{i=1}^{N} [C_{FP} \cdot \mathbb{I}(\hat{y}_i(t) = 1, y_i = 0) + C_{FN} \cdot \mathbb{I}(\hat{y}_i(t) = 0, y_i = 1)] \tag{1}$$

According to Bayesian optimal decision theory, the theoretical optimal threshold under perfect calibration is $t^* = \frac{C_{FP}}{C_{FP}+C_{FN}} = 0.20$.

As shown in Figure 4, under empirical probability density mapping, traversing the probability grid yielded a U-shaped loss curve: when the intervention threshold was locked at $t = 0.7072$, the total social loss reached its empirical minimum (2,401 cost units). The upward shift of the threshold relative to the theoretical value reflects the geometric characteristics of the Random Forest's probability output. A high-sharpness cutoff achieved highly accurate interception of high-risk hosts while strictly controlling false positives.

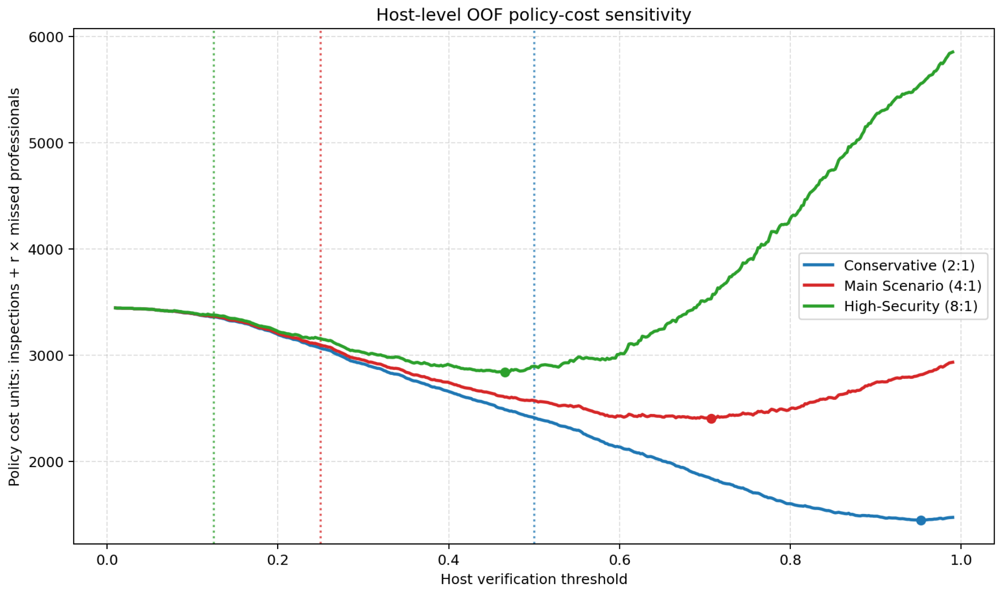

Figure 4: Empirical Cost Curve and Optimal Policy Threshold Determination ($t = 0.7072$)

4 Text as Data: Incremental Value

This section introduces unstructured text such as descriptions (`description`) and amenities (`amenities`) into the model pipeline to quantitatively evaluate the net predictive increment they bring.

4.1 Text Pipeline & Zero-Leakage Extraction

Text processing constructed a dual-branch pipeline. First, the listing descriptions underwent Unigram TF-IDF vectorization (setting $min\_df = 5$ and removing stop words), with the IDF weight calculation formula being:

$$\text{TF-IDF}(t, d, D) = \text{TF}(t, d) \times \left[\log\left(\frac{1 + |D|}{1 + |\{d \in D : t \in d\}|}\right) + 1\right] \tag{2}$$

Second, the JSON amenities list was parsed into a Binary Bag-of-Words (Binary BoW). Text vocabulary construction and weight fitting were strictly confined within the cross-validation training folds. The test set OOV audit showed a median OOV rate of only 2.1%, proving that the in-fold vocabulary has excellent coverage generalizability.

4.2 Predictive Gain & Policy Utility

While keeping the classifier and evaluation environment consistent, the predictive performance of the text-enhanced model achieved a leap.

As shown in Figure 5, after introducing text features (orange line), the test set ROC-AUC jumped from the pure structured baseline of 0.7739 to 0.8766 ($\Delta\text{AUC} = +0.1027$), PR-AUC increased from 0.4356 to 0.6121 ($\Delta\text{AP} = +0.1765$), and Log Loss dropped from 0.4293 to 0.3497. Under a fixed intervention threshold ($t = 0.25$) and the 1:4 cost assumption, text features drastically reduced the number of missed false negatives, causing the global policy loss to plummet from 751 units to 584 units (a drop of 22.2%).

4.3 Proxy Signal Decoding & Limitations

By extracting the text coefficients in the L_2 Logistic Regression, the model precisely restored the micro-picture of Barcelona short-term rentals. Positive coefficients concentrated on `amenity_lock_on_bedroom_door`, HDTV, and shared spaces, revealing the professional host pattern of subdividing homes, independent subletting ("room subdivision"), and standardization. Negative coefficients concentrated on first-person pronouns (`my`, `our`) and words like "quiet", reflecting the emotional narratives of ordinary hosts. An audit

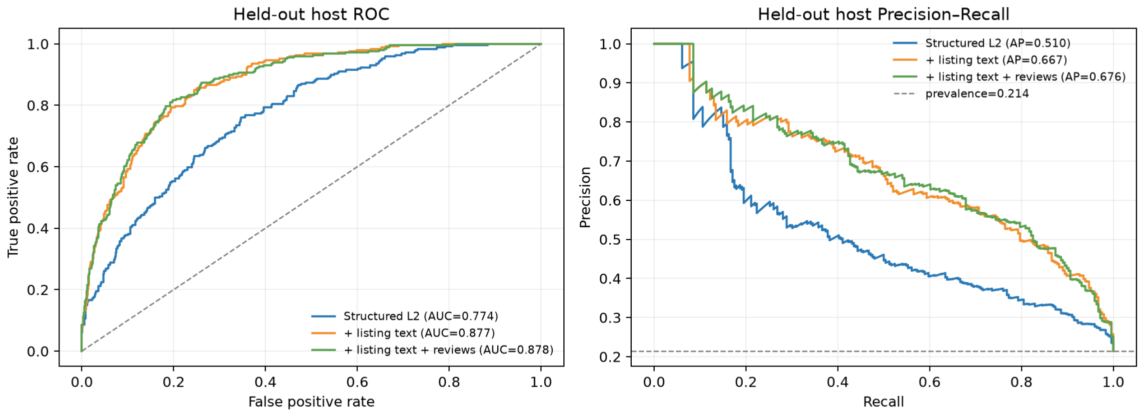

Figure 5: Incremental Predictive Gain from Unstructured Text Features (ROC and PR Curves)

of misclassified cases showed that some ordinary hosts were misjudged due to marketing copywriting, while some professional hosts were missed due to extremely brief text, confirming that text is merely a "proxy signal" rather than a causal essence.

5 Market Structure via Unsupervised Learning

This section departs from supervised labels, employing PCA and K-Means to unsupervisedly deconstruct the geometric structure of the host feature space.

5.1 Host-Level PCA & Variance Pricing

Standardized SVD decomposition was performed on the 12 core features of the 4,595 hosts:

$$X_{\text{scaled}}^T X_{\text{scaled}} v_j = \lambda_j v_j \tag{3}$$

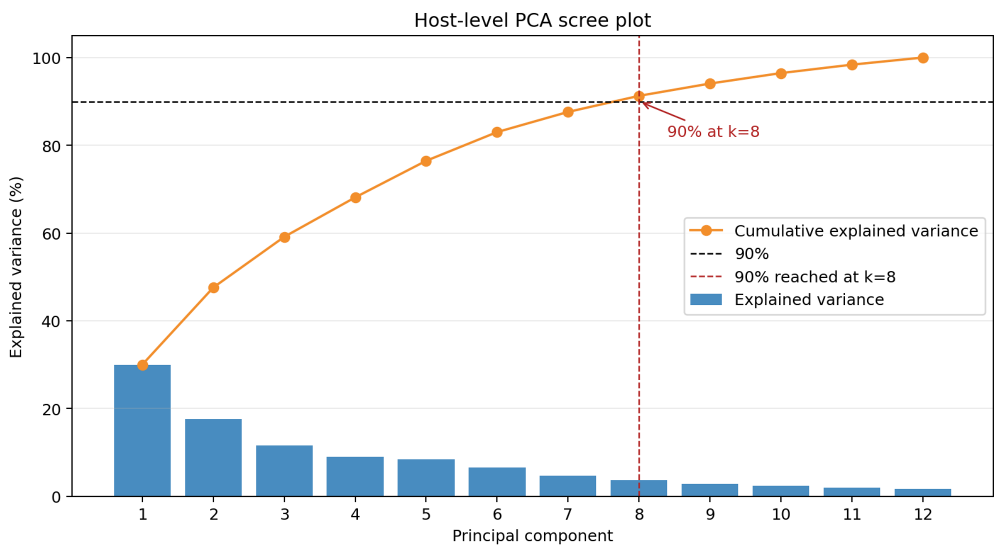

Figure 6: Scree Plot and Cumulative Explained Variance of Principal Components

As shown in the scree plot of Figure 6, the first two principal components explained 30.00% and 17.57% of the variance respectively (cumulative 47.57%). Reaching a 90% cumulative variance required 8 principal components, indicating the high-dimensional complexity of the feature space. The PC1 axis represents asset scale and capital intensity (loadings concentrated on `accommodates` and `log_price_eur`), while the PC2 axis represents calendar restrictions vs. market verification (the opposition of calendar vacancy and historical reviews).

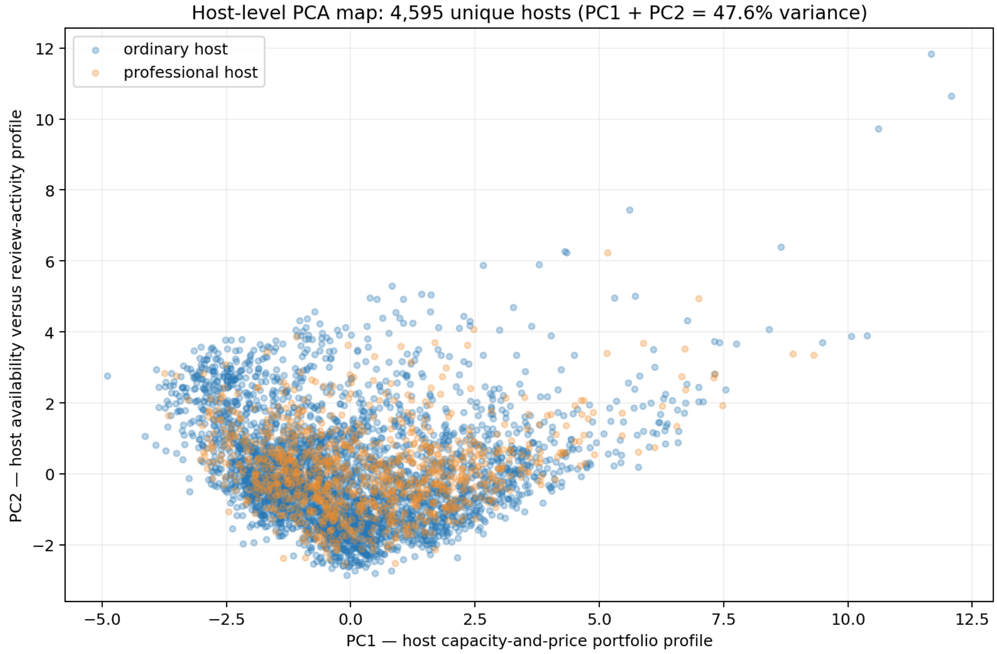

Figure 7: Host-Level PCA Projection Colored by Administrative Binary Target

As shown in Figure 7, projecting the hosts onto the PC1-PC2 map and coloring them with administrative target labels formed a highly overlapping continuous "boomerang" cloud. The linear correlation coefficients were extremely low (0.0833 and 0.0387), indicating that the administratively divided "number of listings" did not naturally segment along the direction of maximum variance.

5.2 K-Means Clustering & Diagnostic Defense

Running the K-Means algorithm on the 12-dimensional space to minimize Inertia: $J(K) = \sum_{k=1}^{K} \sum_{i \in C_k} \|x_i - \mu_k\|^2$. Scanning $K \in [2, 8]$ showed that the average silhouette score peaked at 0.1996 when $K = 2$, locking $K = 2$ as the optimal cut.

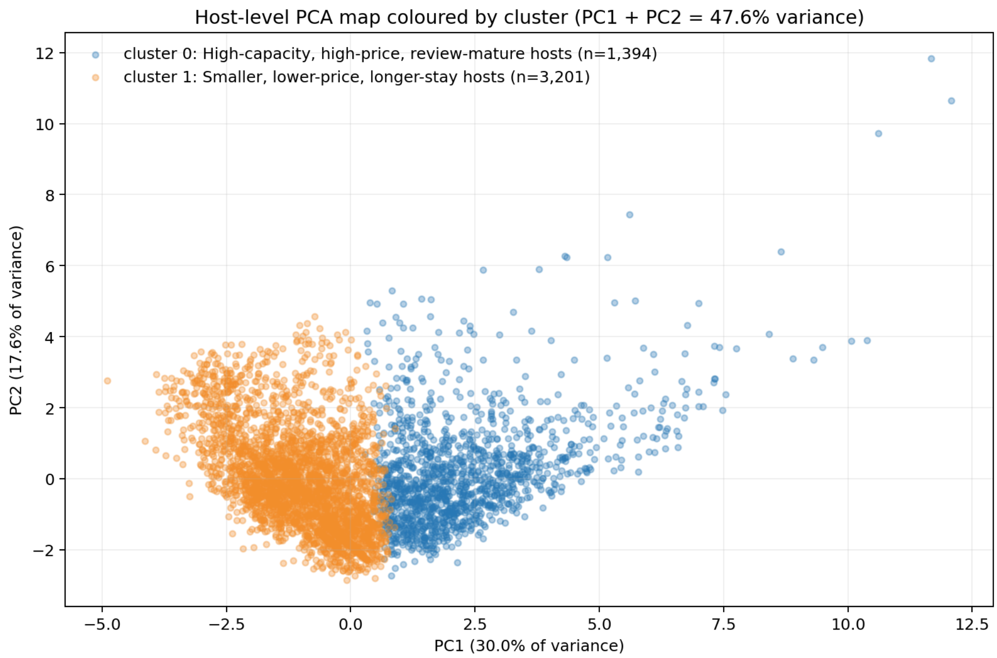

Figure 8: K-Means Clustering of Hosts in PCA Projected Space ($k = 2$)

As shown in Figure 8, the unsupervised K-Means algorithm discarded prior labels and, based on spatial distance, delineated the physical boundary between "high-capacity heavy-asset" (orange area on the right) and "smaller light-asset" (blue area on the left).

5.3 Original-Unit Profile & Operating Models

Restoring to the original business units, Cluster 0 encompassed 1,394 hosts (accounting for 30.34%), accommodating an average of 5.27 people, with an average price of 321.87 EUR, 184.23 reviews, and a minimum stay of only 3.86 days, representing the professional operating matrix of "heavy-asset/high-premium/high-turnover". Cluster 1 encompassed 3,201 hosts (accounting for 69.66%), accommodating an average of 2.33 people, with an average price of 124.03 EUR, and a minimum stay of up to 18.32 days, representing the defensive model of "light-asset/low-ticket/medium-to-long-term rentals".

5.4 External Check & Methodological Findings

Cross-tabulating the clustering labels with the held-out administrative labels revealed a classification recovery rate of only 65.51%, significantly underperforming the majority-class blind guessing baseline (78.65%). In Cluster 0 (heavy-asset high-turnover group), as many as 71.66% (999 hosts) were administratively defined as "ordinary hosts"; meanwhile, in Cluster 1 (light-asset long-term rental group), there lurked 586 "professional hosts". This proves that the administrative binary labels have suffered severe logical disintegration at the level of underlying business models, and quantitative metrics cannot substitute for true operational profiles.

6 Discussion & Limitations

Transforming machine learning models into public policy tools requires extreme epistemological restraint. First, the model captures conditional probability $P(y = 1|X = x)$, not interventional causality $P(y = 1|do(X = x))$; features are merely "behavioral proxy signals". According to Goodhart's Law, if door locks or text features are set as regulatory intervention targets, professional hosts will quickly strategically camouflage themselves. Second, predicted probabilities cannot directly serve as legal evidence for illegal penalties. The correct positioning of the model is as a "Triage Tool" to help inspectors increase clue hit rates; ultimate penalties must rely on offline Human-in-the-Loop verification. Finally, this study is limited by the static nature of single-period cross-sectional data, the vulnerability of public front-end features to adversarial evasion, and the exogenous heuristic assumption of the 1:4 cost matrix.

7 Conclusion

This study established a data-driven framework to identify professional Airbnb hosts in Barcelona using solely public guest-visible information. Under strictly leakage-proof cross-validation, our baseline Random Forest achieved an ROC-AUC of 0.7767. Incorporating unstructured text mining boosted the ROC-AUC to 0.8766 and reduced global policy loss by 22.2%. By applying a 1:4 asymmetric cost matrix, we determined an empirical optimal intervention threshold of $t = 0.7072$, locking the global social loss at a minimum of 2,401 cost units and providing regulators with a quantified standard for law enforcement triage.

Furthermore, unsupervised market deconstruction (PCA and K-Means, $k = 2$) yielded a label recovery rate of only 65.51%, underperforming the majority-class baseline. Crucially, 71.66% of the heavy-asset, high-turnover hosts in Cluster 0 were administratively classified as ordinary hosts. This conclusively proves that the rigid "$\geq$ 3 listings" threshold fails to capture true commercial operational models. Ultimately, future digital urban governance must adhere to a prudent "Human-in-the-Loop" paradigm that combines algorithmic ranking with offline verification.